PROJECT REPORT

Multipurpose Anti-Tamper Box

with RFID Access Control and Environment Monitoring

A low-cost Arduino UNO R3 prototype

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Abstract

This project is a smart storage box that controls who can open it and reports what is happening inside. An RFID card unlocks a servo-driven latch. A Hall effect sensor and a magnet on the lid report whether the lid is open or closed, so the box can tell a legitimate opening from forced entry. A DHT11 sensor measures temperature and humidity, and eight LEDs show the status without any screen. The prototype uses only an Arduino UNO R3, a few low-cost sensors and a cardboard enclosure.

1. Introduction and Problem Statement

Many items need protection from two different problems: unauthorised access and unsuitable conditions. Medicines and vaccines spoil when they get too hot or humid, and they can be misused if anyone can open the box. Valuables, documents, exam papers and laboratory equipment need protection from tampering. Commercial smart lockers and cold-chain monitors exist, but they are expensive and often unavailable where power cuts, heat and limited budgets are common.

The goal of this project is a simple, low-cost box that combines access control, tamper detection and environment monitoring, using only the components available to the team.

2. Objectives

- Allow only authorised RFID cards to unlock the box.
- Detect when the lid is opened without authorisation and raise a tamper alarm.
- Re-lock automatically once the lid has been opened and closed again.
- Monitor temperature and humidity inside the box and warn when limits are crossed.
- Show every state clearly using LEDs, with no display or network needed.
- Use every component in the kit, with each one doing necessary work.

3. Components Used

Component	Qty	Role
Arduino UNO R3	1	Main controller
MFRC522 RFID reader with card or tag	1	Identifies the user
Hall effect sensor with small magnet	1	Detects lid open or closed
DHT11 sensor	1	Measures temperature and humidity
SG90 micro servo	1	Latch that locks and unlocks the lid
LEDs	8	Status display
330 Ω resistors	8	Current limiting for the LEDs
Breadboard and jumper wires	–	Connections
Cardboard box	1	Enclosure

4. System Design

Inputs	Controller	Outputs
RFID reader (card UID)	Arduino UNO R3	SG90 servo (latch)
Hall sensor (lid state)	Compares UID, tracks lock and lid state, and applies temperature and humidity limits	4 temperature LEDs
DHT11 (temperature, humidity)		Humidity, Access, Tamper and All OK LEDs

The RFID reader and the Hall sensor are mounted so they work through or against the enclosure. The RC522 sits flat behind the front panel under a “tap card here” mark. The magnet is fixed to the lid and the Hall sensor to the box body so the two line up when the lid is closed. The DHT11 is placed inside the box so it measures the conditions around the stored items, and the Arduino is kept away from it because the board warms up.

5. Circuit Connections

The RC522 is a 3.3 V device and is the only part powered from the 3.3 V pin. The Hall sensor, DHT11 and servo run from 5 V. All LEDs share one GND rail, each through its own 330 Ω

resistor.

5.1 Module connections

Part	Part pin	UNO pin
RC522	3.3V	3.3V
RC522	GND	GND
RC522	RST	9
RC522	SDA (SS)	10
RC522	MOSI	11
RC522	MISO	12
RC522	SCK	13
Hall sensor	VCC / GND / OUT	5V / GND / 3
DHT11	VCC / GND / DATA	5V / GND / 2
SG90 servo	Red / Brown / Orange	5V / GND / 6

5.2 LED connections

LED	Colour	UNO pin
Temperature 1	Blue	A1
Temperature 2	Green	5
Temperature 3	Yellow	7
Temperature 4	Red	8
High humidity	Yellow	A4
Access granted	Green	4
Tamper	Red	A2
All OK / locked	Blue	A3

The analogue pins A1 to A4 are used as ordinary digital outputs, because the project needs more digital pins than the UNO provides on its own.

6. Working Principle

6.1 Access control

When a card is tapped, the Arduino reads its UID and compares it with the list of authorised cards stored in the code. A valid card moves the servo to the unlocked position. An unknown card leaves the box locked and flashes the red Tamper LED three times.

6.2 Lid sensing and re-locking

The servo is an output only, so it cannot report whether the lid is really closed. The Hall sensor provides that information: when the magnet on the lid is near, the lid is closed. After a valid scan, the box re-locks 2 seconds after the lid has been opened and closed again. If the lid is never opened, it re-locks after 10 seconds.

6.3 Tamper detection

The Arduino knows the box is meant to be locked because it commanded the servo. If the Hall sensor then reports the lid open, the two disagree and the box treats it as forced entry. The red Tamper LED stays on until an authorised card is scanned.

6.4 Environment monitoring

The DHT11 is read every 2 seconds. The four temperature LEDs act as a bar graph. Above the over-temperature limit all four blink together as an alarm. The red humidity LED turns on above the humidity limit.

6.5 Status LEDs

The blue All OK LED is on only when the box is locked, the lid is closed and there is no alarm. The green Access LED is steady while the box is unlocked with the lid closed, and blinks while the lid is open (faster if the lid stays open for more than 30 seconds).

Servo (commanded)	Hall sensor (actual)	Meaning
Locked	Lid closed	Normal, All OK
Locked	Lid open	Forced open, tamper alarm
Unlocked	Lid closed	Valid scan, waiting for the lid to be opened
Unlocked	Lid open	Box in use, Access LED blinks
Unlocked, then lid closed again	Lid closed	Re-lock after 2 seconds

7. Software Design

The programme is written in the Arduino C++ environment and uses the MFRC522, Adafruit DHT and Servo libraries. The main loop runs five small tasks continuously: read the lid sensor with a 50 ms debounce, check for a card, update the lock logic, read the DHT11, and update the LEDs. The loop uses timers (`millis()`) rather than long delays, so the LEDs keep blinking while the box waits for a card. The servo is attached only while it moves, which stops jitter and reduces supply noise.

Setting	Value
Temperature bar LEDs	26, 28, 30 and 32 °C (in the order blue, green, yellow, red)
Blue LED blinks	Below 26 °C
Over-temperature alarm (all temperature LEDs blink)	35 °C and above
High humidity LED	75% and above
Re-lock after lid closes	2 s
Re-lock if the lid is never opened	10 s
Lid left open warning	30 s
Servo to LED delay	1 s

Authorised cards are stored as UID byte arrays. All thresholds, servo angles and the Hall sensor polarity are constants at the top of the sketch, so the behaviour can be tuned without changing the logic.

8. Testing

Each part was tested on its own before the full sketch was assembled.

- **RFID and Hall sensor test:** A combined sketch printed each card UID and the magnet state to the Serial Monitor. Both worked once the wiring was corrected.
- **LED tests:** One sketch lit each LED for 10 seconds in turn, and another lit all eight together. Pins were rearranged after the tests to match the panel layout, and every LED was then confirmed working.
- **Servo test:** The servo was swept between two angles. The direction was reversed, so the locked and unlocked angles were swapped in the code.
- **Full system:** Valid and invalid cards, lid opening and closing, forced opening, and temperature and humidity thresholds were checked against the behaviour described in Section 6.

9. Challenges and Solutions

Challenge	Solution
RFID reader returned firmware 0x0 and a communication failure	Corrected the wiring: 3.3V power and the SPI pins (SDA 10, MOSI 11, MISO 12, SCK 13, RST 9)
Not enough digital pins for 8 LEDs plus four modules	Used A1 to A4 as digital outputs
Servo cannot report whether the lid is closed	Added the Hall sensor, which makes tamper detection possible
Servo turned the wrong way	Swapped the locked and unlocked angles
Servo startup surge can reset the board	Attached the servo only while moving, and a capacitor can be added across its supply
Three red and two green LEDs are easy to confuse	Grouped them on the panel and labelled each one
Hall sensor reacts to only one magnet pole	Fixed the magnet orientation, and made the logic level a setting in the code

10. Applications

- Securing valuables, cash and important documents at home, in shops or in offices.

- Vaccine and medicine storage in clinics, health sub-centres and homes, where heat, humidity and unauthorised access can spoil or lead to misuse of stock.
- Exam paper, evidence or sample boxes that must show if they were opened by an unauthorised person.
- Laboratory or school equipment cabinets that need access control and tamper alerts.
- Food samples, cosmetics, chemicals or electronics that are sensitive to temperature and humidity.

11. Limitations

- The DHT11 is accurate to about ± 2 °C and $\pm 5\%$ humidity, so this is a prototype and not a certified medical cold-chain device.
- The SG90 is a light-duty servo and can only hold a small latch or lid.
- Standard RFID tags can be cloned, so this is not high-security access control.
- The UNO has no Wi-Fi or Bluetooth, so alerts are local (LEDs and the Serial Monitor only).
- A cardboard enclosure does not resist a determined physical attack.

12. Future Scope

- Add a buzzer for audible alarms.
- Add an LCD or OLED to show exact temperature and humidity.
- Move to an ESP32 for Wi-Fi or SMS alerts and remote logging.
- Use a more accurate sensor such as the DS18B20 or DHT22, and a real-time clock for timestamped logs.
- Use a solenoid lock or a stronger servo for heavier doors, and a metal or plastic enclosure.
- Add battery backup so monitoring continues during power cuts.

13. Conclusion

The project shows that a low-cost Arduino system can combine access control, tamper detection and environment monitoring in one box. The key design idea is comparing what the servo was commanded to do with what the Hall sensor actually sees, which turns a simple lock into a tamper-aware one. The prototype met its objectives and works as a base for medicine storage, secure lockers and sensitive-item cabinets.

14. References

1. Arduino Servo library (built in).
2. MFRC522 library by Miguel Balboa.
3. DHT sensor library and Adafruit Unified Sensor library by Adafruit.
4. Datasheets: MFRC522 (NXP), DHT11, SG90 micro servo, and the Hall effect sensor used.